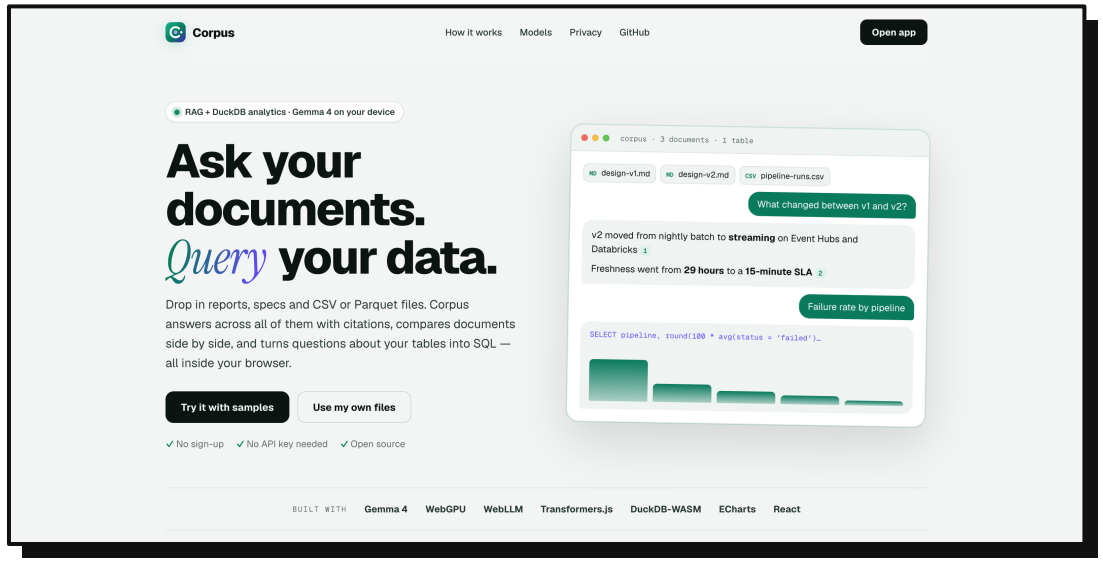
01

CORPUS

Ask your documents. Query your data.

A browser-native RAG and analytics workspace: cited answers across files, side-by-side comparison, and SQL over CSV and Parquet with DuckDB.



GEMMA 4 / WEBGPU / TRANSFORMERS.JS / DUCKDB-WASM / REACT / ECHARTS

02

THE PROBLEM: ANSWERS YOU CANNOT CHECK

Unverifiable answers

Chat tools summarise files fluently, but you cannot see which sentence a claim came from, or whether the newest version won.

Documents and data apart

Design docs and post-mortems live in one tool, the run logs and CSVs in another. Nobody asks both in one place.

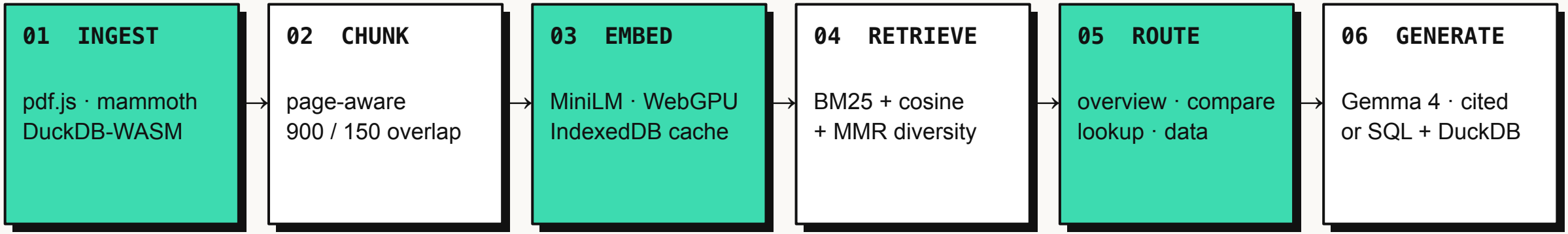
Privacy and cost

Uploading internal files to a hosted LLM is often not allowed, and API keys put a price on every question.

The response: retrieval, SQL and a small model in the browser, every claim cited.

03

ONE PIPELINE, TWO PATHS



Document path

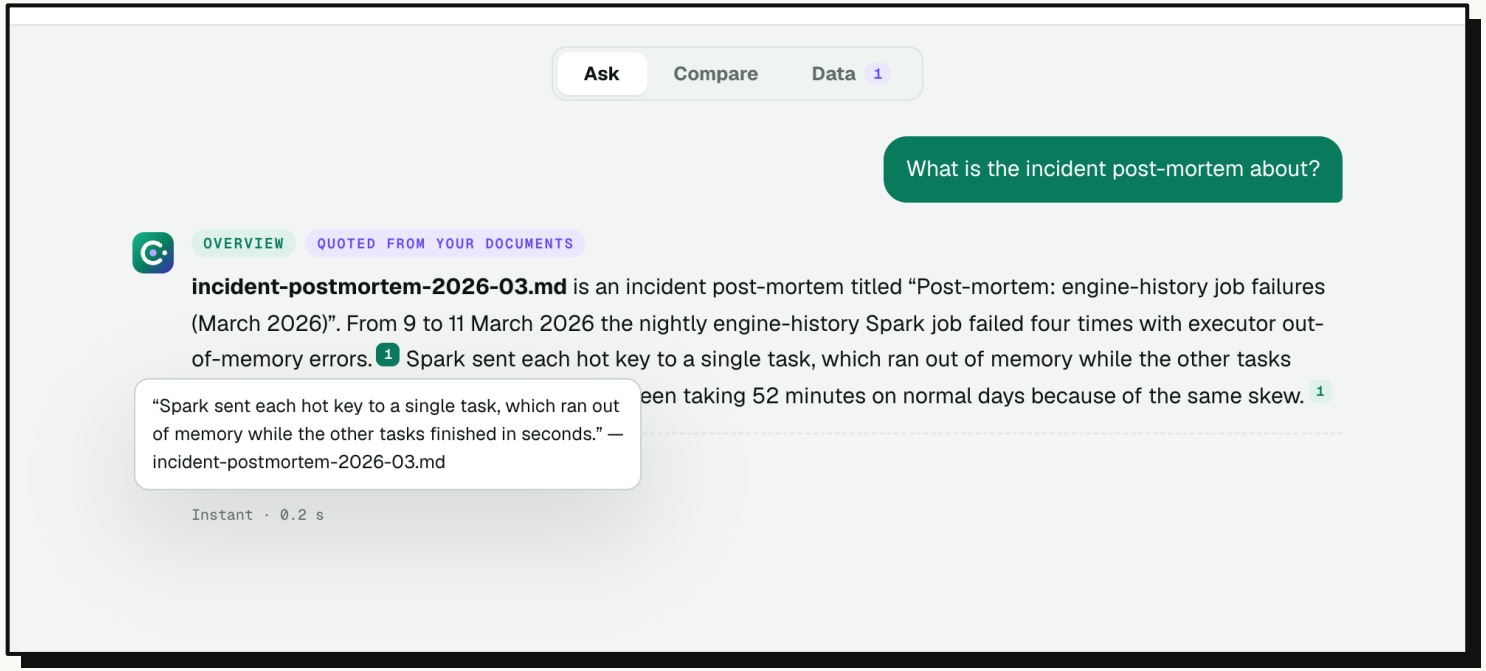
Intent routing picks the strategy: an overview samples the whole file evenly and gets a narrative answer; a comparison retrieves evidence per document; a named file scopes the search.

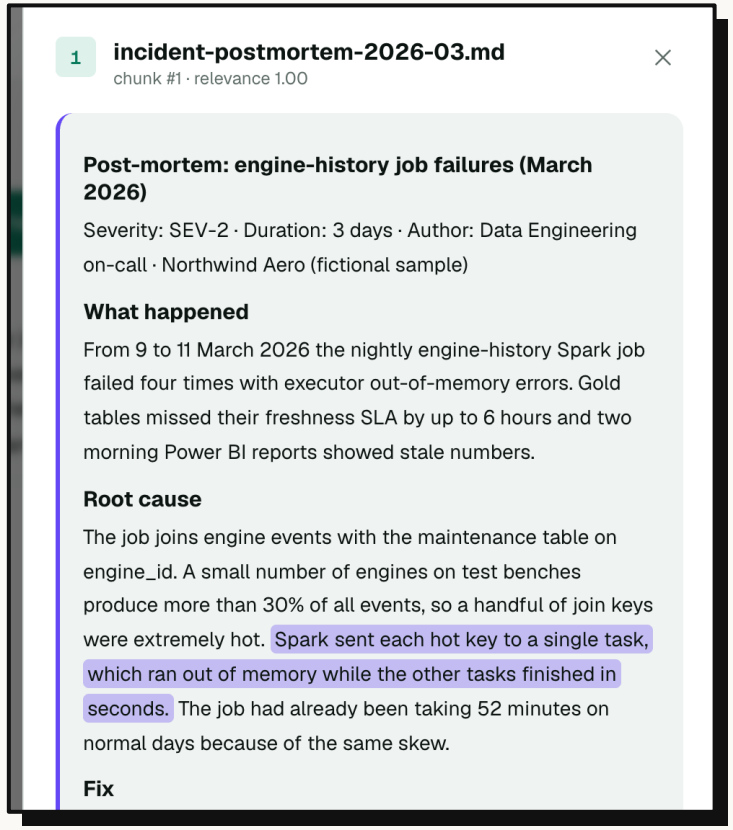
Data path

CSV, Parquet and JSON become DuckDB tables. The table profile is indexed as text, so chat can describe a dataset, and number questions become SQL.

04

EVERY CLAIM CARRIES ITS PROOF

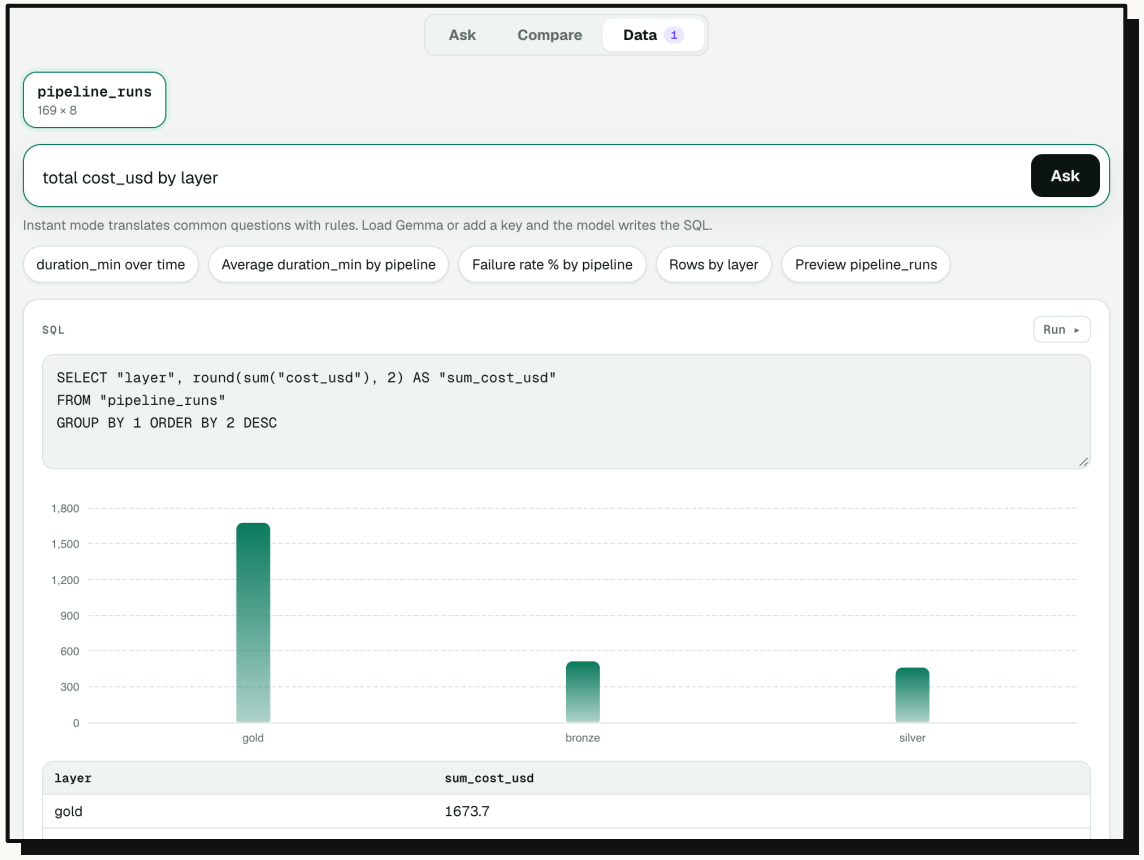




Hover a citation to read the exact supporting sentence; click to open the passage with that sentence highlighted.

05

QUESTIONS BECOME SQL YOU CAN READ



Guarded generation

Only one read-only SELECT runs. A failed query is sent back to the model with the DuckDB error for one corrected retry.

Works without a model

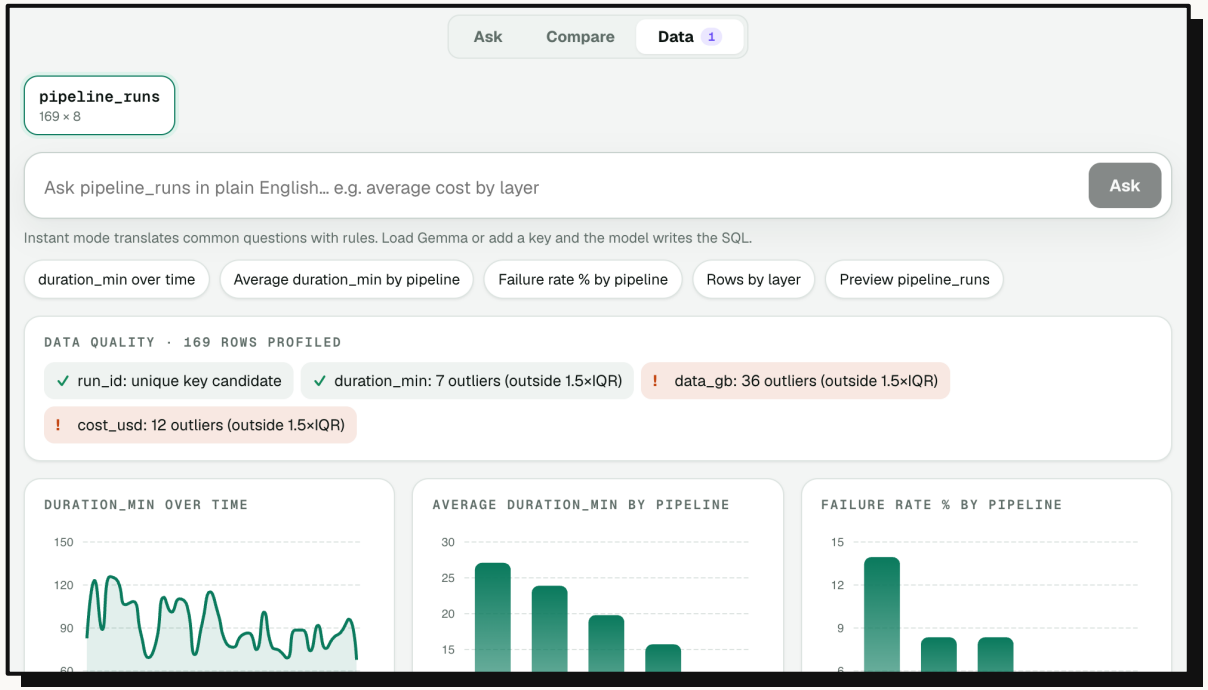
Instant mode translates common questions with rules: "average duration by pipeline", "failed runs over time".

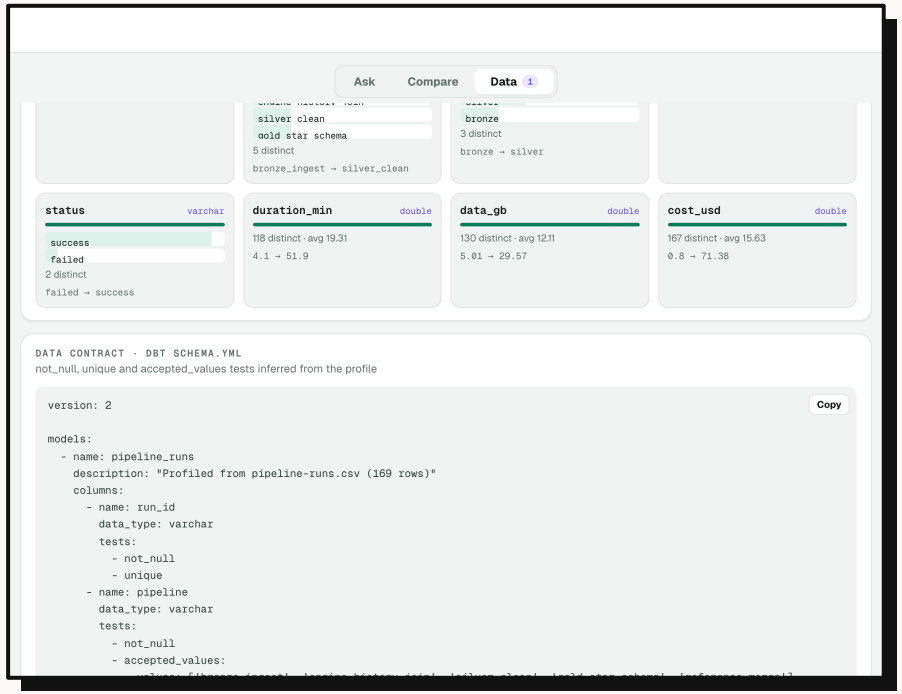
Charted and explained

Results pick a bar or line chart, and the model explains the numbers in one or two plain sentences.

06

PROFILED ON ARRIVAL, SHIPPED AS A CONTRACT





Nulls, cardinality, IQR outliers, duplicates, key candidates

dbt schema.yml: not_null, unique, accepted_values

07

RETRIEVAL IS MEASURED, NOT ASSUMED

RETRIEVER	HIT@1	HIT@3	MRR
Keyword (BM25)	100%	100%	1.00
Vector (cosine)	75%	88%	0.82
Hybrid + MMR	**100%**	**100%**	**1.00**

Eight golden questions over the sample documents, scored live in the pipeline panel. Keywords catch exact terms; vectors catch meaning; the hybrid keeps both.

Small-model accuracy

On-device models get a compressed context (the best sentences per passage) plus a citation safety net.

Zero re-work on refresh

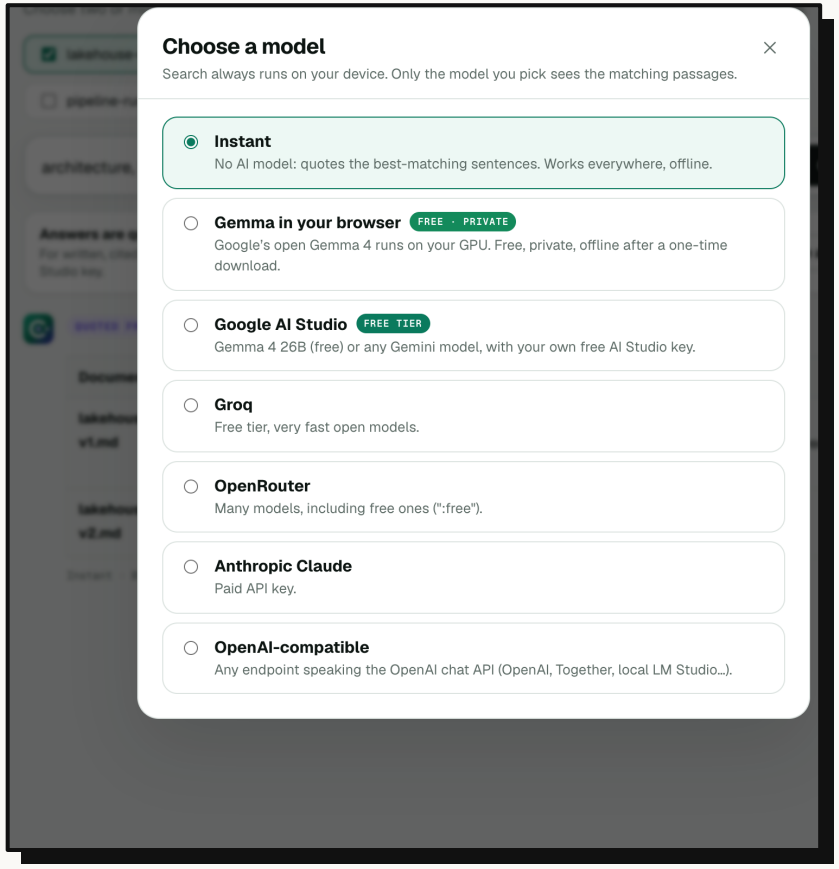
Documents, table files and every embedding persist in IndexedDB, keyed by content hash.

GPU where it helps

Embeddings run on WebGPU in a worker (fp16), falling back to WASM; cached per text.

08

FREE TO RUN, PRIVATE BY DEFAULT



Gemma 4 in the browser

Gemma 4 E2B on WebGPU via WebLLM. Free and offline after a one-time 2.8 GB download.

Bring your own key

Google AI Studio (Gemma 4 26B free tier), Groq, OpenRouter, Anthropic or any OpenAI-compatible API. Keys stay in the browser.

Instant mode

No model at all: ranked, cited quotes and rule-based SQL in milliseconds.

adityamalviya04692.github.io/Project-Corpus/

github.com/adityamalviya04692/Project-Corpus